

Simulation-based Inference with the Python Package sbijax

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

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Editor: 

Submitted: 19 March 2026

Published: unpublished

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Summary

Neural simulation-based inference (SBI) describes an emerging family of methods for Bayesian inference for simulator models that use neural networks as surrogate models. Here we introduce sbijax, a Python package that implements a wide variety of state-of-the-art methods in neural simulation-based inference using a user-friendly programming interface. sbijax offers high-level functionality to quickly construct SBI estimators, and compute and visualize posterior distributions with only a few lines of code. In addition, the package provides functionality for conventional approximate Bayesian computation, to compute model diagnostics, and to automatically estimate summary statistics. By virtue of being entirely written in JAX, sbijax is extremely computationally efficient, allowing rapid training of neural networks and executing code automatically in parallel on both CPU and GPU.

Statement of Need

Modern approaches to neural simulation-based inference (SBI) utilize recent developments in neural density estimation or score-based generative modelling to build surrogate models to approximate Bayesian posterior distributions. Similarly to conventional methods, such as approximate Bayesian computation (ABC) and its sequential (SMC-ABC) and annealing-based (e.g., SABC) variants, neural SBI methods infer this posterior distribution by first simulating synthetic data and then numerically constructing an appropriate approximation to this pseudo data set. SBI methods are attractive for a couple of reasons. On the one hand, this family of methods has been shown to be more computationally efficient and often more accurate than ABC methods, in particular for smaller simulation budgets. On the other hand, SBI allows to easily amortize inference, i.e., to infer the posterior distribution for multiple different observations once a neural model has been trained.

Here we propose sbijax, a Python package implementing state-of-the-art methodology of neural simulation-based inference. While the main focus of the package is the implementation of recent algorithms to make them available to practitioners, e.g., Wildberger et al. (2023) or Schmitt et al. (2023), sbijax also implements common methods from approximate Bayesian computation, e.g., SMC-ABC (Beaumont et al., 2009), to have the entire SBI toolbox in one efficient package (see Table 1 for an overview). In addition, sbijax provides functionality for model diagnostics, posterior visualization and Markov Chain Monte Carlo (MCMC) sampling.

The package uses the high-performance computing framework JAX as a backend (Bradbury et al., 2018). Using JAX has several advantages, including a) that it uses the same syntax as numpy (Harris et al., 2020) which enables a seamless transition for applied scientists who already are familiar with it, and b) that empirical evaluations have shown that JAX can be significantly faster than PyTorch (see, e.g., Phan et al. (2019)). Our package heavily builds on

libraries from the JAX-verse and common Bayesian inference tools. Specifically, we use Haiku (Hennigan et al., 2020) to construct and train neural networks, ArviZ (Kumar et al., 2019) for various model visualizations, surjectors for normalizing-flow based density estimation (Dirmeier, 2024), TensorFlow Probability (Dillon et al., 2017) to define statistical distributions, and BlackJAX (Cabezas et al., 2024) for posterior sampling using Markov Chain Monte Carlo.

Table 1: Implemented SBI methods in sbijax .

Model	Class name	Reference
Sequential Monte Carlo ABC	SMCABC	Beaumont et al. (2009)
Neural likelihood estimation	NLE	Papamakarios et al. (2019)
Surjective neural likelihood estimation	SNLE	Dirmeier et al. (2025)
Automatic posterior transformation	NPE	Greenberg et al. (2019)
Contrastive neural ratio estimation	NRE	Miller et al. (2022)
Flow matching posterior estimation	FMPE	Wildberger et al. (2023)
Posterior Score Estimation	NPSE	Sharrock et al. (2024)
All-In-One Posterior Estimation	AI0	Gloeckler et al. (2024)
Consistency model posterior estimation	CMPE	Schmitt et al. (2023)
Neural approximate sufficient statistics	NASS	Chen et al. (2021)
Neural approximate slice sufficient statistics	NASSS	Chen et al. (2023)

State of the field

While a plethora of different models has been proposed in the recent literature, the development of adequate software packages has not followed at the same pace, and only few packages exist that allow modelers to use these methods. Most prominently, the Python package `sbi` (Tejero-Cantero et al., 2020) implements several approaches for neural simulation-based inference, such as a neural posterior, likelihood-ratio, and likelihood estimation (Cranmer et al., 2020) utilizing a PyTorch backend (Paszke et al., 2019). The package additionally provides an API for model diagnostics, such as posterior predictive checks, effective sample size computations and simulation-based calibration. However, the package lacks implementations of recent developments which pose the state-of-the-art in the field, such as by Chen et al. (2023), Dirmeier et al. (2025) or Schmitt et al. (2023). Also, by virtue of being developed in PyTorch it is potentially restrictive to practitioners that do not have experience with it. For approximate Bayesian computation, several Python packages are available. In particular `abcpy` (Dutta et al., 2021) implements a multitude of different ABC algorithms. However, none of these packages implement modern (neural) SBI methods.

Software design

Sbijax is designed as a modular and extensible toolbox for SBI in JAX, combining a functional programming philosophy with a flexible interface for both expert users and domain scientists. Its design is guided by the following principles:

- 1) Faithful alignment with JAX's functional paradigm. Sbijax adheres closely to the low-level, functional programming model of JAX. While SBI methods are implemented using an object-oriented structure, all member functions return immutable objects rather than modifying internal state. This avoids hidden side effects, facilitates composability, and ensures compatibility with JAX transformations such as `jit`, `vmap`, and `pmap`.
- 2) Separation of inference methods and neural network implementations. Sbijax focuses on implementing SBI algorithms, while neural network components are defined using Haiku. This allows users to construct models directly with Haiku, and to incorporate

probabilistic building blocks from Distrax (DeepMind et al., 2020) and surjectors (Dirmeier, 2024). As a result, model definitions remain flexible and low-level, while seamlessly interoperating with the broader JAX ecosystem.

3) Support for extensibility and research. Sbijax is structured to facilitate experimentation with new SBI methods. Its modular design allows components such as neural architectures, training objectives, and sampling strategies to be easily replaced or extended, making it suitable as both a research framework and a practical toolbox.

4) Accessibility for domain scientists. In addition to its flexibility, Sbijax includes pre-implemented models with sensible defaults, enabling use without in-depth expertise in deep learning. In the simplest case, users only need to define a prior and a simulator to run inference workflows—often in as few as five lines of code.

Research impact statement

sbijax has already been used extensively in the Machine Learning literature. Dirmeier et al. (2025) and Dirmeier & Mira (2025) proposed novel SBI methods for surjective neural likelihood estimation and posterior estimation using causal constraints, respectively, where they used sbijax heavily for their experimental evaluations. Ulzega et al. (2025) used sbijax to infer the posterior distribution of a complicated Bayesian model from the astrophysics literature. Albert et al. (2025) developed a novel ABC method that uses sbijax for model evaluation.

AI usage disclosure

No GenAI or other AI tools have been used in writing the software or this manuscript.

Acknowledgements

This research was supported by the Swiss National Science Foundation (Grant No. 200021_208249).

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